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Semiconductor package

Abstract

A semiconductor package includes a semiconductor chip comprising an active surface and an inactive surface facing each other. At least one antenna module is arranged adjacent to the semiconductor chip. The at least one antenna module comprises a main antenna and a sub-antenna. A redistribution structure is disposed on the semiconductor chip and the at least one antenna module. The redistribution structure electrically connects the active surface of the semiconductor chip to the at least one antenna module. A molding member surrounds the semiconductor chip and the at least one antenna module. The inactive surface of the semiconductor chip and the main antenna are exposed from the molding member, and the sub-antenna is covered by the molding member.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8269337	12/2011	Hu et al.	N/A	N/A
8736031	12/2013	Lee et al.	N/A	N/A
9196951	12/2014	Baks et al.	N/A	N/A
9337526	12/2015	Hong et al.	N/A	N/A
9711465	12/2016	Liao et al.	N/A	N/A
9721920	12/2016	Beer et al.	N/A	N/A
9991216	12/2017	Liao et al.	N/A	N/A
10368442	12/2018	Yu et al.	N/A	N/A
10461399	12/2018	Ndip et al.	N/A	N/A
10700024	12/2019	So et al.	N/A	N/A
10790244	12/2019	Huang et al.	N/A	N/A
2015/0070228	12/2014	Gu	343/727	H01Q 25/00
2015/0280327	12/2014	Spella	N/A	N/A
2016/0056544	12/2015	Garcia	343/725	H01Q 9/0407
2016/0126634	12/2015	Liu et al.	N/A	N/A
2016/0178730	12/2015	Trotta	342/175	G01S 13/931
2018/0191052	12/2017	Ndip	N/A	H01L 24/20
2021/0384645	12/2020	Jia	N/A	H01Q 1/243
2022/0247065	12/2021	Zhu	N/A	H01L 24/16
2023/0420396	12/2022	Acikalin	N/A	H01L 23/66

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
110943054	12/2019	CN	H01L 21/56
10-1208241	12/2011	KR	N/A

OTHER PUBLICATIONS

Andreas Stelzer et al., “Integrated Microwave Sensors in SiGe with Antenna in Package:”, ICECOM(2016), 8 pages. cited by applicant

C.-T, Wang et al., “InFO_AIP Technology for High Performance and Compact 5G Millimeter Wave System Integration”, ECTC(2018), pp. 202-207. cited by applicant

Guochi Huang et al., “A low cost 60GHz antenna in Fan-Out Panel Level Package for millimeter-wave radar application”, ICEPT(2020), 4 pages. cited by applicant

M. PourMousavi et al., “Antenna Design and Characterization for a 61 GHz Transceiver in eWLB Package”, European Microwave Integrated Circuit Conference (2013), pp. 1415-1418. cited by applicant

Imran Aziz et al., “Broadband Fan-out Phased Antenna Array at 28 GHz for 5G Applications”, EUMC(2020), pp. 212-215. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2021-0064222, filed on May 18, 2021 in the Korean Intellectual Property Office, the disclosure of which is incorporated by reference in its entirety herein.

1. TECHNICAL FIELD

(2) The present inventive concept relates to a semiconductor package, and more particularly, to a semiconductor package including an antenna module.

2. DISCUSSION OF RELATED ART

(3) As the manufacturing technology and wireless communication technology of a semiconductor package has increased, a semiconductor package having an integrated antenna has been developed. For example, a method of forming an antenna pattern in a redistribution layer of a semiconductor package has been developed. However, providing a sufficient radiation performance of the antenna may result in a constraint in design of the semiconductor package or the occurrence of performance errors. Accordingly, a semiconductor chip is desired in which a semiconductor chip and the antenna module are stably integrated so that a degree of freedom of design is increased and the occurrence of performance error is reduced or prevented.

SUMMARY

(4) An embodiment of the present inventive concept provides a semiconductor package in which a semiconductor chip and the antenna module are stably integrated so that a degree of freedom of design is increased and a performance error is reduced.

(5) The issues to be solved by embodiments of the present inventive concept are not limited to the above-mentioned issues, and other issues not mentioned may be clearly understood by those of ordinary skill in the art from the following description.

(6) According to an embodiment of the present inventive concept, a semiconductor package includes a semiconductor chip comprising an active surface and an inactive surface facing each

other. At least one antenna module is arranged adjacent to the semiconductor chip. The at least one antenna module comprises a main antenna and a sub-antenna. A redistribution structure is disposed on the semiconductor chip and the at least one antenna module. The redistribution structure electrically connects the active surface of the semiconductor chip to the at least one antenna module. A molding member surrounds the semiconductor chip and the at least one antenna module. The inactive surface of the semiconductor chip and the main antenna are exposed from the molding member, and the sub-antenna is covered by the molding member.

(7) According to an embodiment of the present inventive concept, a semiconductor package includes a semiconductor chip comprising an active surface and an inactive surface facing each other. At least one antenna module is arranged adjacent to the semiconductor chip. The at least one antenna module comprises a main antenna and a sub-antenna. A redistribution structure is disposed on the semiconductor chip and the at least one antenna module. The redistribution structure electrically connects the active surface of the semiconductor chip to the at least one antenna module. A molding member surrounds the semiconductor chip and the at least one antenna module. A heat discharging member is arranged on the semiconductor chip. The main antenna is exposed from the molding member, and the sub-antenna is covered by the molding member. A thickness of the at least one antenna module is greater than a thickness of the semiconductor chip.

(8) According to an embodiment of the present inventive concept, a semiconductor package includes a semiconductor chip comprising an active surface and an inactive surface facing each other. A molding member comprises an opening in a portion thereof, and surrounding the semiconductor chip. At least one antenna module is arranged in the opening of the molding member. The at least one antenna module comprises a main antenna and a sub-antenna on a printed circuit board. A redistribution structure is arranged under the semiconductor chip, the molding member, and the at least one antenna module. The redistribution structure electrically connects the active surface of the semiconductor chip to the at least one antenna module. An external connection member is arranged under the redistribution structure. The main antenna is exposed to the outside from the molding member, and the sub-antenna directly contacts the molding member.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Embodiments of the present inventive concept will be more clearly understood from the following detailed description taken in conjunction with the accompanying drawings in which:
- (2) FIG. 1A is a plan view of a semiconductor package according to an embodiment of the present inventive concept;
- (3) FIG. 1B is a cross-sectional view taken along line B-B' of FIG. 1A according to an embodiment of the present inventive concept;
- (4) FIG. 1C is an enlarged view of region CC in FIG. 1B according to an embodiment of the present inventive concept;
- (5) FIG. 2 is a plan view of a semiconductor package according to an embodiment of the present inventive concept;
- (6) FIG. 3A is a plan view of a semiconductor package according to an embodiment of the present inventive concept;
- (7) FIG. 3B is a cross-sectional view taken along line B-B' of FIG. 3A according to an embodiment of the present inventive concept;
- (8) FIG. 4 is a plan view of a semiconductor package according to an embodiment of the present inventive concept;
- (9) FIG. 5 is a block diagram of a manufacturing method of a semiconductor package, according to an embodiment of the present inventive concept;

(10) FIGS. **6** through **11** are cross-sectional views illustrating a manufacturing method of a semiconductor package, according to embodiments of the present inventive concept; and
(11) FIG. **12** is a configuration diagram of an electronic device including a semiconductor package, according to an embodiment according to an embodiment of the present inventive concept.

DETAILED DESCRIPTION OF EMBODIMENTS

(12) Hereinafter, embodiments of the present inventive concept are described in detail with reference to the accompanying drawings.

(13) FIG. **1A** is a plan view of a semiconductor package **10** according to an embodiment, FIG. **1B** is a cross-sectional view taken along line B-B' in FIG. **1A**, and FIG. **1C** is an enlarged view of region CC in FIG. **1B**.

(14) Referring to FIGS. **1A** through **1C**, the semiconductor package **10** of the present inventive concept may include a semiconductor chip **100**, an antenna module **200** including a main antenna **201** and a sub-antenna **202**, a molding member **300**, a redistribution structure **400**, and an external connection member **500**.

(15) The semiconductor chip **100** may be used in an electronic device including a semiconductor element. The semiconductor chip **100** may include a semiconductor substrate **101** having an active surface **101A** and an inactive surface **101B** facing each other. For example, in an embodiment, the active surface **101A** and the inactive surface **101B** may be spaced apart from each other in the Z direction which is a thickness direction of the semiconductor chip **100**. In an embodiment, a circuit unit for implementing an integrated circuit function of the semiconductor chip **100** may be formed on the active surface **101A** of the semiconductor substrate **101** by using a semiconductor manufacturing process. For example, in an embodiment, an individual unit element, a metal layer, an interlayer insulating layer **110**, and an electrode pad **120** may be formed on the active surface **101A** of the semiconductor substrate **101**.

(16) In an embodiment, the semiconductor chip **100** may include a logic chip or a memory chip. The logic chip may include, for example, a microprocessor, an analog element, or a digital signal processor. In addition, the memory chip may include, for example, a volatile memory such as dynamic random access memory (RAM) (DRAM) and static RAM (SRAM), or a non-volatile memory such as phase-change RAM (PRAM), magnetoresistive RAM (MRAM), resistive RAM (RRAM), and ferroelectric RAM (FRAM). In an embodiment, the semiconductor chip **100** may include a high bandwidth memory.

(17) In an embodiment, the semiconductor chip **100** may be a communication chip that includes a signal processing circuit for processing a wireless signal. For example, the semiconductor chip **100** may include a radio-frequency integrated circuit. For example, the semiconductor chip **100** may include a radio-frequency integrated circuit for a millimeter wave (mmWave) having a wavelength in a range of about 1 mm to about 10 mm and a wavelength close thereto. The semiconductor chip **100** may include a radio-frequency integrated circuit for a frequency band in a range of about 28 GHz to about 300 GHz and a frequency band adjacent thereto.

(18) In addition, in other embodiments, the semiconductor chip **100** may have a chiplet structure including a plurality of semiconductors.

(19) In an embodiment, the antenna module **200** may include the main antenna **201**, the sub-antenna **202**, a base board **210**, and a wiring layer **220**. A thickness **200T** (e.g., length in the Z direction) of the antenna module **200** may be substantially the same as a thickness **100T** (e.g., length in the Z direction) of the semiconductor chip **100**.

(20) In an embodiment, the main antenna **201** may include a patch antenna. The patch antenna includes a plurality of parts, which are separated from each other. For example, as shown in an embodiment of FIG. **1A**, the patch antenna may include a plurality of patches that are arranged in a plurality of rows extending in the X direction and columns extending in the Y direction in the uppermost surface **200S** of the antenna module **200**. In addition, at least a portion of the main antenna **201** may be arranged to be exposed to the outside. For example, as shown in FIGS. **1B-1C**,

in an embodiment, an upper surface of the main antenna **201** may be exposed to the outside. In an embodiment, the main antenna **201** may include, for example, a director, a radiator, and a reflector. The main antenna **201** may transceive a wireless signal of mmWave.

(21) In an embodiment, the sub-antenna **202** may include a Yagi antenna. In an embodiment, the sub-antenna **202** may be arranged at a certain interval (e.g., a first interval) on one side surface of the antenna module **200**, such as one lateral side surface of the antenna module **200**. The sub-antenna **202** may include, for example, a director, a projector, and a reflector. The sub-antenna **202** may transceive a wireless signal of mmWave.

(22) In an embodiment, each of the main antenna **201** and the sub-antenna **202** may include, for example, electrolytically deposited (ED) copper, a rolled-annealed (RA) copper foil, a stainless steel foil, an aluminum foil, an ultra-thin copper foil, sputtered copper, copper alloys, nickel, stainless steel, beryllium copper, etc. However, embodiments of the present inventive concept are not limited thereto.

(23) The base board **210** may include a printed circuit board (PCB). However, embodiments of the present inventive concept are not limited thereto and the base board **210** may vary. In an embodiment, the base board **210** may include at least one material selected from phenol resin, epoxy resin, and polyimide. In an embodiment, the base board **210** may include at least one material selected from FR4, tetrafunctional epoxy, polyphenylene ether, epoxy/polyphenylene oxide, bismaleimide triazine, thermount, cyanate ester, polyimide, and liquid crystal polymer. However, embodiments of the present inventive concept are not limited thereto.

(24) The wiring layer **220** may penetrate the base board **210** and electrically connect the main antenna **201** and the sub-antenna **202** to the redistribution structure **400**. For example, in an embodiment, the wiring layer **220** may extend from an upper surface of the antenna module **200** to a lower surface of the antenna module **200**. In an embodiment, the wiring layer **220** may include, sputtered copper, copper alloys, nickel, stainless steel, beryllium copper, etc. The wiring layer **220** may include a plurality of layers. However, embodiments of the present inventive concept are not limited thereto and the material of the wiring layer **220** and the number of layers of the wiring layer **220** may vary. In an embodiment, the wiring layer **220** may include a first wiring layer electrically connecting the main antenna **201** to the redistribution structure **400** and a second wiring layer electrically connecting the sub-antenna **202** to the redistribution structure **400**.

(25) As shown in FIG. 1C, in some embodiments, the antenna module **200** may further include an antenna protection layer **230** covering an upper surface of the main antenna **201**. The antenna protection layer **230** may function as an antioxidant layer preventing oxidation of the main antenna **201**. In an embodiment, the antenna protection layer **230** may include, for example, polyimide, polyester, photoimageable coverlay (PIC), a photo-imageable solder resist, an organic solderability preservative, etc. However, embodiments of the present inventive concept are not limited thereto.

(26) The molding member **300** may protect the semiconductor chip **100** and the antenna module **200** from external influences such as contamination and impact. In an embodiment, the molding member **300** may include an epoxy mold compound, resin, etc. In addition, the molding member **300** may be formed by a process such as compression molding, lamination, and screen printing. In some embodiments, the molding member **300** may be formed to expose the inactive surface **101B** of the semiconductor chip **100** and the uppermost surface **200S** of the antenna module **200** to the outside. At least a portion of the main antenna **201** may be exposed from the molding member **300**, and the sub-antenna **202** may be arranged to be covered by the molding member **300** and the sub-antenna **202** directly contacts the molding member **300**. For example, as shown in the embodiment of FIG. 1B, an upper surface of the main antenna **201** may be exposed by the molding member **300**. These may be characteristics of the semiconductor package **10** according to an embodiment of the present inventive concept, and detailed descriptions thereof will be given later.

(27) The molding member **300** may constitute an outer shape of the semiconductor package **10**, and the redistribution structure **400** may be arranged by using the molding member **300**. As shown in

FIG. 1B, in an embodiment, the redistribution structure **400** may be disposed on lower surfaces of the semiconductor chip **100**, the antenna module **200** and the molding member **300**. In an embodiment, the thickness **100T** (e.g., length in the Z direction) of the semiconductor chip **100**, the thickness **200T** (e.g., length in the Z direction) of the antenna module **200**, and a thickness **300T** (e.g., length in the Z direction) of the molding member **300** may be substantially the same as each other. For example, a level (e.g., length in the Z direction from an upper surface of the redistribution structure **400**) of the uppermost surface the semiconductor chip **100** (e.g., a level of an upper surface of the inactive surface **101B**), a level of the uppermost surface **200S** of the antenna module **200**, and a level of the uppermost surface **300S** of the molding member **300** may be substantially the same as each other (e.g., substantially co-planar in the Z direction).

(28) The redistribution structure **400** may include an insulating layer **410** and a metal distribution layer **420**. In an embodiment, the redistribution structure **400** may include copper (Cu), nickel (Ni), gold (Au), chromium (Cr), titanium (Ti), or palladium (Pd), or may include the metal distribution layer **420** including an alloy thereof. The metal distribution layer **420** may be formed by using an electroplating process. However, embodiments of the present inventive concept are not limited thereto and the composition of the redistribution structure **400** and the method of forming the redistribution structure **400** may vary.

(29) The metal distribution layer **420** of the redistribution structure **400** may be arranged to electrically connect the electrode pad **120** of the semiconductor chip **100** to the wiring layer **220** of the antenna module **200**. For example, the metal distribution layer **420** may connect a portion of the electrode pad **120**, which processes a communication signal, to the wiring layer **220**.

(30) In addition, the metal distribution layer **420** of the redistribution structure **400** may be arranged to electrically connect the electrode pad **120** of the semiconductor chip **100** to the external connection member **500**. For example, the metal distribution layer **420** may connect the other portion of the electrode pad **120**, which functions as a ground, to the external connection member **500**.

(31) In an embodiment, the external connection member **500** may include a solder ball or a solder bump. In some embodiments, a material constituting the external connection member **500** may include a lead free solder including tin (Sn). The semiconductor package **10** may be connected to an external device, such as a main board, etc., via the external connection member **500**.

(32) Electronic devices have become increasingly miniaturized and multi-functional according to the rapid development of the electronics industry and demands of users. Accordingly, the semiconductor chip **100** used in an electronic device has become increasingly miniaturized and multi-functional. Accordingly, an interval of the electrode pad **120** included in the semiconductor package **10** has been increasingly reduced.

(33) When the number of electrode pads **120** for miniaturization or inputs/outputs of the semiconductor chip **100** are increased, the semiconductor package **10** may not accommodate all signal terminals in the semiconductor chip **100**. Accordingly, the semiconductor package **10** may extend the redistribution structure **400** to the outside of the semiconductor chip **100**, and may expand a region in which the electrode pad **120** is arranged. For example, in some embodiments, a fan-out wafer level package (FO-WLP), or a fan-out panel level package (FO-PLP) (hereinafter, both are referred to as the FO-WLP) may be applied to the semiconductor package **10**.

(34) Accordingly, in an embodiment, the semiconductor package **10** having a general FO-WLP structure may arrange the external connection member **500** on an expanded surface of the semiconductor package **10** by forming the redistribution structure **400** on the electrode pad **120**, and a position of the electrode pad **120** and a position where the external connection member **500** is formed may be varied by using the redistribution structure **400**.

(35) However, as packaging technology and wireless communication technology have advanced, a technology of integrating an antenna inside the semiconductor package has been developed. In some embodiments, a method of forming an antenna pattern in a redistribution layer of a

semiconductor package is employed. In this embodiment, since the redistribution layer used as an antenna is not exposed to the outside to secure radiation performance of the antenna, there may be a constraint of the design, or there may be an increased occurrence of performance errors.

(36) However, the semiconductor package **10** according to the present inventive concept may insert the antenna module **200** into the molding member **300** and expose the main antenna **201** to the outside (e.g., to the air) in the FO-WLP structure to reduce signal loss due to impedance matching.

(37) In addition, the semiconductor package **10** according to an embodiment of the present inventive concept may separate the semiconductor chip **100** and the antenna module **200** into completely independent bodies, and may reduce interference between the main antenna **201** and the sub-antenna **202**, and the semiconductor chip **100**.

(38) In addition, the semiconductor package **10** according to the present inventive concept may reduce constraints of the design (e.g., a thickness and a size of a patch) of the main antenna **201**, due to the design rule of the redistribution structure **400**.

(39) Therefore, the semiconductor package **10** according to an embodiment of the present inventive concept may increase the degree of freedom of design and decrease the occurrence of a performance error by stably integrating the semiconductor chip **100** and the antenna module **200**.

(40) FIG. 2 is a plan view of a semiconductor package **20** according to an embodiment of the present inventive concept.

(41) Most of the components constituting the semiconductor package **20** and materials included in the components to be described below may be substantially the same as or similar to those described with reference to FIGS. 1A through 1C. Thus, for convenience of description, differences from the semiconductor package **10** described above are mainly described and a repeated description of similar or identical components may be omitted.

(42) Referring to FIG. 2, the semiconductor package **20** may include the semiconductor chip **100**, first to third antenna modules **200A**, **200B**, and **200C**, the molding member **300**, a redistribution structure, and an external connection member.

(43) The first to third antenna modules **200A**, **200B**, and **200C** may each be constituted by a plurality of parts, which are separated from each other, and each of the plurality of parts may be arranged to be spaced apart by a certain interval from any side surface of the semiconductor chip **100**. For example, the plurality of parts may form a group, and form the first to third antenna modules **200A**, **200B**, and **200C**. However, embodiments of the present inventive concept are not limited thereto and the number of the plurality of antenna modules may vary.

(44) Each of the plurality of parts of the first to third antenna modules **200A**, **200B**, and **200C** may include the main antenna **201**, the sub-antenna **202**, a base board, and a wiring layer. In some embodiments, each of the plurality of parts of the first to third antenna modules **200A**, **200B**, and **200C** may include a PCB. In an embodiment, a thickness (e.g., length in the Z direction) of each of the plurality of parts of the first to third antenna modules **200A**, **200B**, and **200C** may be substantially the same as the thickness **100T** of the semiconductor chip **100**.

(45) FIG. 3A is a plan view of a semiconductor package **30** according to an embodiment of the present inventive concept, and FIG. 3B is a cross-sectional view taken along line B-B' in FIG. 3A.

(46) Most of components constituting the semiconductor package **30** and materials included in the components to be described below may be substantially the same as or similar to those described with reference to FIGS. 1A through 1C. Thus, for convenience of description, differences from the semiconductor package **10** described above are mainly described and a repeated description of similar or identical components may be omitted.

(47) Referring to FIGS. 3A and 3B together, the semiconductor package **30** according to an embodiment of the present inventive concept may include the semiconductor chip **100**, an antenna module **200'** including the main antenna **201** and the sub-antenna **202**, the molding member **300**, the redistribution structure **400**, the external connection member **500**, and a heat discharging member **600**.

(48) The heat discharging member **600** may include a conductive material. For example, in an embodiment, the heat discharging member **600** may include at least one metal selected from Cu, Al, Ti, Ta, tungsten (W), Ni, or may include a metal paste including the metal, or may include a metal tape including the metal. However, embodiments of the present inventive concept are not limited thereto.

(49) Heat generated by the semiconductor chip **100** via the heat discharging member **600** may be discharged to the outside of the semiconductor package **10**. As shown in FIG. **3B**, the heat discharging member **600** may be exposed by the molding member **300**. For example, the molding member **300** may not cover the heat discharging member **600**. In an embodiment, the heat discharging member **600** may be disposed on an upper surface of the semiconductor chip **100** and a portion of the upper surface of the molding member **300** and the lateral side surfaces and upper surface of the heat discharging member **600** may be exposed. For example, the inactive surface **101B** may directly contact a lower surface of the heat discharging member **600**.

(50) The heat discharging member **600** and the molding member **300** may constitute an external shape of the semiconductor package **30**, and the redistribution structure **400** may be arranged by using the molding member **300**. In an embodiment, due to the existence of the heat discharging member **600**, the thickness **100T** of the semiconductor chip **100** and the thickness **300T** of the molding member **300** may be substantially the same as each other. However, a thickness **200'T** of the antenna module **200'** may be greater than the thickness **100T** of the semiconductor chip **100**. For example, the level of the uppermost surface of the semiconductor chip **100** (e.g., the inactive surface **101B**) and the level of the uppermost surface **300S** of the molding member **300** may be substantially the same as each other, and a level of an uppermost surface **200'S** of the antenna module **200'** may be higher than these levels. In some embodiments, the level of the uppermost surface **200'S** of the antenna module **200'** may be higher than a level of a lowermost surface of the heat discharging member **600**, and may be lower than the level of the uppermost surface of the heat discharging member **600**. As shown in FIG. **3B**, in an embodiment, a portion of a side surface of the antenna module **200'** and an upper surface of the antenna module **200'** may be exposed from the molding member **300**.

(51) FIG. **4** is a plan view of a semiconductor package **40** according to an embodiment of the present inventive concept.

(52) Most of the components constituting the semiconductor package **40** and materials included in the components to be described below may be substantially the same as or similar to those described with reference to FIGS. **1A** through **1C**, **3A**, and **3B**. Thus, for convenience of description, differences from the semiconductor packages **10** and **30** described above are mainly described and a repeated description of similar or identical components may be omitted.

(53) Referring to FIG. **4**, the semiconductor package **30** may include the semiconductor chip **100**, first to third antenna modules **200'A**, **200'B**, and **200'C**, the molding member **300**, a redistribution structure, and an external connection member, and the heat discharging member **600**.

(54) The first to third antenna modules **200'A**, **200'B**, and **200'C** may be constituted by a plurality of parts, which are separated from each other, and each of the plurality of parts may be arranged to be spaced apart by a certain interval (e.g., a first interval) from any side surface of the semiconductor chip **100**. For example, the plurality of parts may form a group, and form the first to third antenna modules **200'A**, **200'B**, and **200'C**.

(55) Each of the plurality of parts of the first to third antenna modules **200'A**, **200'B**, and **200'C** may include the main antenna **201**, the sub-antenna **202**, a base board, and a wiring layer. In some embodiments, each of the plurality of parts of the first to third antenna modules **200'A**, **200'B**, and **200'C** may include a PCB.

(56) Due to the existence of the heat discharging member **600**, the thickness **100T** of the semiconductor chip **100** and the thickness **300T** of the molding member **300** may be substantially the same as each other, but a thickness of each of the plurality of parts of the first to third antenna

modules **200'A**, **200'B**, and **200'C** may be greater than the thickness **100T** of the semiconductor chip **100**. For example, the level of the uppermost surface of the semiconductor chip **100** and the level of the uppermost surface of the molding member **300** may be substantially the same as each other, and a level of each of the plurality of parts of the first to third antenna modules **200'A**, **200'B**, and **200'C** may be higher than these levels. In some embodiments, the level of the uppermost surface of each of the plurality of parts of the antenna modules **200'A**, **200'B**, and **200'C** may be higher than the level of the lowermost surface of the heat discharging member **600**, and may be lower than the level of the uppermost surface of the heat discharging member **600**.

(57) FIG. 5 is a block diagram of a manufacturing method **S10** of a semiconductor package, according to an embodiment of the present inventive concept.

(58) Referring to FIG. 5, the manufacturing method **S10** of a semiconductor package may include a process sequence of first through seventh operations in blocks **S110** through **S170**.

(59) When a certain embodiment is implemented differently, a particular process sequence may be executed differently from the sequence to be described. For example, two consecutively described processes may be substantially performed at the same time or in a sequence opposite to the described sequence.

(60) The manufacturing method **S10** of a semiconductor package according to an embodiment of the present inventive concept may include the first operation in block **S110** in which a semiconductor chip is mounted on a first carrier substrate, the second operation in block **S120** in which an antenna module is mounted on the first carrier substrate, the third operation in block **S130** in which a preliminary molding member covering both the semiconductor chip and the antenna module is formed, the fourth operation in block **S140** in which a second carrier substrate is attached and the first carrier substrate is removed, the fifth operation in block **S150** in which the redistribution structure is formed, the sixth operation in block **S160** in which the second carrier substrate is removed and the preliminary molding member is manufactured as the molding member, and the seventh operation in block **S170** in which the external connection member is attached to the redistribution structure.

(61) Technical characteristics of each of the first through seventh operations in blocks **S110** through **S170** are described in detail later with reference to FIGS. 6 through 11.

(62) FIGS. 6 through 11 are cross-sectional views illustrating a manufacturing method of a semiconductor package **10**, according to embodiments of the present inventive concept.

(63) Referring to FIG. 6, the semiconductor chip **100** may be mounted on a first carrier substrate **CS1**.

(64) In an embodiment, the first carrier substrate **CS1** may include, for example, glass, silicon, or aluminum oxide. However, embodiments of the present inventive concept are not limited thereto. In an embodiment, an adhesive layer may be formed between the first carrier substrate **CS1** and the semiconductor chip **100** (e.g., in the Z direction) to attach the semiconductor chip **100** to the first carrier substrate **CS1**. However, embodiments of the present inventive concept are not limited thereto. The adhesive layer may have a liquid shape or a gel shape so that a shape of the adhesive layer is easily changeable under a certain level of pressure.

(65) The semiconductor chip **100** may include a semiconductor substrate **101** which includes the active surface **101A** and the inactive surface **101B** facing each other (e.g., in the Z direction).

(66) In an embodiment, the semiconductor substrate **101** may include, for example, silicon. However, embodiments of the present inventive concept are not limited thereto. For example, in an embodiment, the semiconductor substrate **101** may include a semiconductor element such as germanium (Ge), or a compound semiconductor such as silicon carbide (SiC), gallium arsenide (GaAs), indium arsenide (InAs), and indium phosphide (InP). In an embodiment, the semiconductor substrate **101** may have a silicon on insulator (SOI) structure. For example, the semiconductor substrate **101** may include a buried oxide (BOX) layer. The semiconductor substrate **101** may include a conductive region, for example, a well doped with impurities, or a structure

doped with impurities. In addition, the semiconductor substrate **101** may have various device isolation structures such as a shallow trench isolation (STI) structure.

(67) In an embodiment, a circuit unit including an individual unit element for implementing an integrated circuit function of a semiconductor element by using the semiconductor manufacturing process may be formed in the semiconductor substrate **101**.

(68) On the semiconductor substrate **101**, the interlayer insulating layer **110** arranged between a metal layer and the semiconductor substrate **101** may be formed. In an embodiment, the interlayer insulating layer **110** may include a low-k material layer having a lower permittivity than silicon oxide. For example, a dielectric material constituting the interlayer insulating layer **110** may include oxides such as phosphor silicate glass (PSG), boro-phosphor silicate glass (BPSG), undoped silicate glass (USG), tetra ethyl ortho silicate (TEOS), plasma enhanced-TEOS (PE-TEOS), high density plasma-chemical vapor deposition (HDP-CVD) oxide, a low dielectric material used in back end of line (BEOL), an ultra-low dielectric material, etc. However, embodiments of the present inventive concept are not limited thereto.

(69) In some embodiments, the interlayer insulating layer **110** may be formed in a structure in which a first interlayer insulating layer, a second interlayer insulating layer, a third interlayer insulating layer, and a fourth interlayer insulating layer are sequentially stacked (e.g., in the Z direction). However, embodiments of the present inventive concept are not limited thereto and the number of interlayer insulating layers constituting the interlayer insulating layer **110** may vary.

(70) The electrode pad **120** may be electrically connected to a circuit unit of the semiconductor chip **100**, and may electrically connect the semiconductor chip **100** to an external electrical device. The electrode pad **120** may, as a portion via which an electrical signal is input/output to/from the semiconductor chip **100**, be equipped in a plurality on the semiconductor chip **100**, and may include Al, W, Cu, Ni, or a combination thereof. However, embodiments of the present inventive concept are not limited thereto. In an embodiment, after a metal such as Al is formed at a certain thickness on the semiconductor chip **100**, the electrode pad **120** may be manufactured by patterning a shape of the electrode pad **120** by using a photo processing and an etching process. However, embodiments of the present inventive concept are not limited thereto and the method in which the electrode pad **120** is formed may vary.

(71) Referring to FIG. 7, the antenna module **200** may be mounted on the first carrier substrate CS1 in parallel with the semiconductor chip **100**.

(72) The antenna module **200** may be spaced apart by a certain interval (e.g., a first interval) from one side surface of the semiconductor chip **100**, and may be attached to the first carrier substrate CS1. For example, in an embodiment, the antenna module **200** may be spaced apart from a lateral side surface of the semiconductor chip **100** by a first interval in the X direction. In an embodiment, an adhesive layer may be formed between the first carrier substrate CS1 and the antenna module **200** to attach the antenna module **200** to the first carrier substrate CS1. The adhesive layer may have a liquid shape or a gel shape so that a shape of the adhesive layer is easily changeable under certain pressure. However, embodiments of the present inventive concept are not limited thereto.

(73) The antenna module **200** may include the main antenna **201**, the sub-antenna **202**, the base board **210**, and the wiring layer **220**. In an embodiment, the main antenna **201** may include a patch antenna. In addition, the main antenna **201** may be arranged to be exposed to the outside. For example, an upper surface of the main antenna **201** may be exposed to the outside. The sub-antenna **202** may include a Yagi antenna. The sub-antenna **202** may be arranged at a certain interval on one side surface of the antenna module **200**.

(74) Referring to FIG. 8, a preliminary molding member **300P** covering both the semiconductor chip **100** and the antenna module **200** may be formed.

(75) The preliminary molding member **300P** may protect the semiconductor chip **100** and the antenna module **200** from external influences such as impact. In an embodiment, the preliminary molding member **300P** may include an epoxy mold compound (EMC), resin, etc. In addition, the

preliminary molding member **300P** may be formed by using processes such as compression molding, lamination, and screen printing. However, embodiments of the present inventive concept are not limited thereto.

(76) In some embodiments, the preliminary molding member **300P** may be formed to cover both an upper surface and lateral side surfaces of each of the semiconductor chip **100** and the antenna module **200**. In other embodiments, the preliminary molding member **300P** may surround only lateral side surfaces of each of the semiconductor chip **100** and the antenna module **200** so that the upper surface of each of the semiconductor chip **100** and the antenna module **200** is exposed to the outside. In addition, in some embodiments, the preliminary molding member **300P** may be formed before the antenna module **200** is attached to the first carrier substrate **CS1**. In this embodiment, the preliminary molding member **300P** may be formed to include an opening where the antenna module **200** is arranged in a subsequent process.

(77) Referring to FIG. **9**, a second carrier substrate **CS2** may be attached on the preliminary molding member **300P** so that the second carrier substrate **CS2** faces the first carrier substrate (**CS1**, refer to FIG. **8**). For example, the second carrier substrate **CS2** may be disposed on an opposite surface (e.g., in the Z direction) of the preliminary molding member **300P** as compared to the first carrier substrate **CS1**.

(78) To remove the first carrier substrate (**CS1**, refer to FIG. **8**) and perform a subsequent process, the second carrier substrate **CS2** may be attached. In an embodiment, the second carrier substrate **CS2** may include, for example, glass, silicon, or aluminum oxide. However, embodiments of the present inventive concept are not limited thereto. In an embodiment, an adhesive layer may be formed between the second carrier substrate **CS2** and the preliminary molding member **300P** (e.g., in the Z direction) to attach the preliminary molding member **300P** to the second carrier substrate **CS2**.

(79) Thereafter, the first carrier substrate (**CS1**, refer to FIG. **8**) may be removed. After the first carrier substrate (**CS1**, refer to FIG. **8**) is removed, the second carrier substrate **CS2** may be overturned and a subsequent process may be performed so that the second carrier substrate **CS2** is positioned under the preliminary molding member **300P**.

(80) Referring to FIG. **10**, the redistribution structure **400** may be formed on the semiconductor chip **100**, the antenna module **200**, and the preliminary molding member **300P**.

(81) The redistribution structure **400** may include an insulating layer **410** and the metal distribution layer **420**. For example, in an embodiment, the redistribution structure **400** may include Cu, Ni, Au, Cr, Ti, or Pd, or may include the metal distribution layer **420** including an alloy thereof. The metal distribution layer **420** may be formed by using an electroplating process. However, embodiments of the present inventive concept are not limited thereto.

(82) The metal distribution layer **420** of the redistribution structure **400** may be arranged to electrically connect the electrode pad **120** of the semiconductor chip **100** to the wiring layer **220** of the antenna module **200**. For example, the metal distribution layer **420** may connect a portion of the electrode pad **120**, which processes a communication signal, to the wiring layer **220**.

(83) Referring to FIG. **11**, the second carrier substrate (**CS2**, refer to FIG. **10**) may be removed, and the preliminary molding member (**300P**, refer to FIG. **10**) may be manufactured as the molding member **300**.

(84) In an embodiment, a laser may be irradiated to the second carrier substrate (**CS2**, refer to FIG. **10**) to separate/remove the second carrier substrate (**CS2**, refer to FIG. **10**). Due to irradiation of the laser, a bonding force between the preliminary molding member (**300P**, refer to FIG. **10**) and the second carrier substrate (**CS2**, refer to FIG. **10**) may be weakened.

(85) In an embodiment, the molding member **300** may then be manufactured by grinding the preliminary molding member (**300P**, refer to FIG. **10**) by using a chemical mechanical polishing (CMP) process or an etch-back process so that the inactive surface **101B** of the semiconductor chip **100** and the main antenna **201** of the antenna module **200** are exposed. In this manner, the thickness

100T of the semiconductor chip **100**, the thickness **200T** of the antenna module **200**, and the thickness **300T** of the molding member **300** may be substantially the same as each other.

(86) Referring to FIG. **1B** again, the external connection member **500** may be attached to the redistribution structure **400**, such as a lower surface of the redistribution structure **400**.

(87) In some embodiments, the external connection member **500** may include a solder ball. The solder ball may be formed in a sphere shape, and may be attached to the redistribution structure **400**. In other embodiments, the external connection member **500** may be formed as a solder bump on the redistribution structure **400**. However, embodiments of the present inventive concept are not limited thereto.

(88) By using the manufacturing process described above, the semiconductor package **10** according to an embodiment of the present inventive concept may be completed. While the semiconductor package **10** has been described with a chip-first manufacturing method of the FO-WLP, embodiments of the present inventive concept are not limited thereto. For example, the semiconductor package **10** may be implemented in a chip-last manufacturing method in which the redistribution structure **400** is formed on the first carrier substrate (**CS1**, refer to FIG. **6**) in advance, and the semiconductor chip **100** is then mounted on the redistribution structure **400**.

(89) Therefore, the semiconductor package **10** according to an embodiment of the present inventive concept may, by stably integrating the semiconductor chip **100** and the antenna module **200**, increase the degree of freedom of design, and decrease the occurrence of performance errors.

(90) FIG. **12** is a configuration diagram of an electronic device **1000** including a semiconductor package, according to an embodiment of the present inventive concept.

(91) Referring to FIG. **12**, the electronic device **1000** may include a main board **1010**. The main board **1010** may include a chipset **1020**, a network **1030**, an other components **1040**, or the like that may be physically and/or electrically connected to each other. These components may be combined with other electronic components to be described later to form various types of signal lines **1090**.

(92) In an embodiment, the chipset **1020** may include a memory chip such as a volatile memory, a non-volatile memory, and a flash memory, an application processor chip such as a central processor, a graphics processor, a digital signal processor, an encoding processor, a microprocessor, and a microcontroller, and a logic chip such as an analog to digital converter and an application-specific integrated circuit (ASIC), etc. In addition, other types of chip-related electronic components may be included. In addition, the chipset **1020** may be combined with each other.

(93) In an embodiment, the network **1030** may include Wi-Fi (IEEE 802.11 family, or the like), WiMAX (IEEE 802.16 family, or the like), IEEE 802.20, long term evolution (LTE) (IEEE), EV-DO, HSPA+, HSDPA+, HSUPA+, EDGE, GSM, GPS, GPRS, CDMA, TDMA, DECT, Bluetooth, 3G, 4G, 5G, and any other wired/wireless protocols designated subsequently. In addition, any other wired/wireless standards and protocols may be included. In addition, the network **1030** may be combined with each other with the chipset **1020**.

(94) In an embodiment, the other components **1040** may include a high frequency inductor, a ferrite inductor, a power inductor, a ferrite bead, low temperature co-firing ceramics (LTCC), an electromagnetic interference (EMI) filter, a multi-layer ceramic capacitor (MLCC), etc. In addition, manual components used for other various applications, or the like, may be included. In addition, the other components **1040** may be combined with each other with the chipset **1020** and/or the network **1030**.

(95) Depending on a type of electronic device **1000**, the electronic device **1000** may include other electronic components which are physically and/or electrically connected to the main board **1010**, or are not so. In an embodiment, other electronic components, may include, for example, a camera **1050**, an antenna **1060**, a display **1070**, a battery **1080**, an audio codec, a video codec, a power amplifier, a compass, an accelerometer, a gyroscope, a speaker, a bulk storage, a compact disk (CD), a digital versatile disk (DVD), etc. In addition, depending on the type of electronic device **1000**, other electronic components used for various applications, or the like, may be included.

(96) In an embodiment, the electronic device **1000** may include a smartphone, a personal digital assistant, a digital video camera, a digital still camera, a network system, a computer, a monitor, a tablet, a laptop, a netbook, a television, a video game console, a smart watch, a smart watch, etc. In addition, the electronic device **1000** may include any other electronic device processing data. For example, the electronic device **1000** may be various different small, medium or large-scaled electronic devices.

(97) A semiconductor package according to an embodiment of the present inventive concept may be applied to various applications such as the electronic device **1000** which varies as described above. For example, the semiconductor packages **10**, **20**, **30**, and **40** described above with reference to FIGS. **1A** through **4** may be mounted in the electronic device **1000** as applications of antennas and memory chips.

(98) While the present inventive concept has been particularly shown and described with reference to non-limiting embodiments thereof, it will be understood that various changes in form and details may be made therein without departing from the spirit and scope of the present inventive concept.

Claims

1. A semiconductor package comprising: a semiconductor chip comprising an active surface and an inactive surface facing each other; at least one antenna module arranged adjacent to the semiconductor chip, the at least one antenna module comprising a main antenna and a sub-antenna; a redistribution structure disposed on the semiconductor chip and the at least one antenna module, the redistribution structure electrically connecting the active surface of the semiconductor chip to the at least one antenna module; and a molding member surrounding the semiconductor chip and the at least one antenna module, wherein the molding member the semiconductor chip and the main antenna are co-planar with each other in a thickness direction of the semiconductor package, wherein the inactive surface of the semiconductor chip and the main antenna are exposed from the molding member, and the sub-antenna is covered by the molding member.
2. The semiconductor package of claim 1, wherein: the at least one antenna module comprises a printed circuit board; and the printed circuit board comprises a plurality of wiring layers electrically connecting each of the main antenna and the sub-antenna to the redistribution structure.
3. The semiconductor package of claim 2, further comprising an antenna protection layer disposed on the main antenna.
4. The semiconductor package of claim 1, wherein a thickness of the at least one antenna module is greater than or equal to a thickness of the semiconductor chip.
5. The semiconductor package of claim 1, wherein: the at least one antenna module includes a plurality of parts that are spaced apart from each other; and the plurality of parts are spaced apart from one side surface of the semiconductor chip by a first interval.
6. The semiconductor package of claim 1, wherein a level of an uppermost surface of the semiconductor chip, a level of an uppermost surface of the at least one antenna module, and a level of an uppermost surface of the molding member are substantially identical to each other.
7. The semiconductor package of claim 1, wherein: the main antenna comprises a patch antenna arranged on an upper surface of the antenna module; and the sub-antenna comprises a Yagi antenna arranged on a side surface of the antenna module.
8. The semiconductor package of claim 7, wherein, in a planar view, patches of the main antenna are arranged in a plurality of rows and columns.
9. The semiconductor package of claim 1, wherein, in the semiconductor package, the main antenna is exposed to the outside.
10. The semiconductor package of claim 1, further comprising an external connection member arranged under the redistribution structure, wherein the semiconductor package comprises a fan-out wafer level package.

11. A semiconductor package comprising: a semiconductor chip comprising an active surface and an inactive surface facing each other; at least one antenna module arranged adjacent to the semiconductor chip, the at least one antenna module comprising a main antenna and a sub-antenna, the main antenna and the sub-antenna are disposed on a same layer as each other, a redistribution structure disposed on the semiconductor chip and the at least one antenna module, the redistribution structure electrically connecting the active surface of the semiconductor chip to the at least one antenna module; a molding member surrounding the semiconductor chip and the at least one antenna module, wherein the molding member, the semiconductor chip and the at least one antenna module are co-planar with each other in a thickness direction of the semiconductor package; and a heat discharging member arranged on the semiconductor chip, wherein the main antenna is exposed from the molding member, and the sub-antenna is covered by the molding member, and a thickness of the at least one antenna module is greater than a thickness of the semiconductor chip.

12. The semiconductor package of claim 11, wherein: the inactive surface of the semiconductor chip directly contacts a lower surface of the heat discharging member; a level of an upper surface of the at least one antenna module is higher than a level of the lower surface of the heat discharging member; and the level of the upper surface of the at least one antenna module is lower than a level of an upper surface of the heat discharging member.

13. The semiconductor package of claim 12, wherein a portion of a side surface of the at least one antenna module is exposed from the molding member.

14. The semiconductor package of claim 11, wherein the at least one antenna module comprises a printed circuit board, wherein the printed circuit board comprises: a first wiring layer electrically connecting the main antenna to the redistribution structure; and a second wiring layer electrically connecting the sub-antenna to the redistribution structure.

15. The semiconductor package of claim 11, wherein: the main antenna comprises a patch antenna arranged on an upper surface of the at least one antenna module; the sub-antenna comprises a Yagi antenna arranged on a side surface of the antenna module; and the semiconductor package comprises a fan-out wafer level package.

16. A semiconductor package comprising: a semiconductor chip comprising an active surface and an inactive surface facing each other; a molding member comprising an opening in a portion thereof, and surrounding the semiconductor chip; at least one antenna module arranged in the opening of the molding member, the at least one antenna module comprising a main antenna and a sub-antenna on a printed circuit board, wherein the molding member, the semiconductor chip and the main antenna are co-planar with each other in a thickness direction of the semiconductor package; a redistribution structure arranged under the semiconductor chip, the molding member, and the at least one antenna module, the redistribution structure electrically connecting the active surface of the semiconductor chip to the at least one antenna module; and an external connection member arranged under the redistribution structure, wherein the main antenna is exposed to the outside from the molding member, and the sub-antenna directly contacts the molding member.

17. The semiconductor package of claim 16, wherein the printed circuit board comprises: a base board; and a wiring layer configured to penetrate the base board, the wiring layer electrically connecting the redistribution structure to the main antenna, and the redistribution structure to the sub-antenna.

18. The semiconductor package of claim 16, wherein: the main antenna comprises a patch antenna having a plurality of rows and columns and arranged on an upper surface of the antenna module, the sub-antenna comprises a Yagi antenna arranged on a side surface of the antenna module, and the semiconductor package comprises a fan-out wafer level package.

19. The semiconductor package of claim 18, wherein: the patch antenna includes a plurality of parts that are separated from each other; and the plurality of parts are spaced apart from one side surface of the semiconductor chip by a first interval.

20. The semiconductor package of claim 16, further comprising a heat discharging member arranged to directly contact the inactive surface of the semiconductor chip.
